

1 - INTRODUCTION

Hot melt extrusion (HME), adopted from the plastics industry, is gaining increasing importance in pharmaceutical product development due to several advantages. Firstly, it is a continuous process with few steps which allows straightforward control of process parameters. Secondly, extrudates can be processed into a variety of dosage forms. The HME technique has proven beneficial when poorly soluble drugs need to be dissolved in thermoplastic polymers to improve solubility and bioavailability. Alternatively, HME can be applied to formulate water soluble compounds into release retarding matrices, which provide controlled drug delivery [1]. Several lipid excipients have been investigated for this approach including waxes, (partial) glycerides and vegetable oils [2]. However, little is known about the process boundaries for lipid excipients. In the present work the importance of process parameters and features of lipid excipients were evaluated using Compritol® 888 ATO (a glyceryl behenate with low HLB (1) and high melting point (70°C)) as the matrix former.

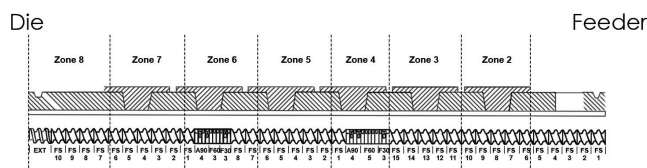
2 - MATERIALS & METHODS

Materials

Glyceryl behenate (Compritol® 888 ATO, Gattefosse, France); hydrophilic dye (FDC, bleu N1 W092, Wackherr, France); lipophilic dye (vert au gras W7212, Wackherr); niacin (Sigma).

Hot melt extrusion

Extrusion of Compritol® 888 was conducted using a twin screw extruder (Pharma 11, Thermo Fisher, Germany) and the following basic screw design:



Drug distribution and homogeneity in the lipophilic Compritol® 888 matrix was simulated by manually adding 3mg of a hydrophilic or lipophilic dye. Impact and limitations of feed rate, screw speed and temperature on Compritol® 888 extrusion was investigated by modifying these parameters step wise. Following the evaluation of "placebo" formulations, niacin was added at a fixed concentration of 10% to evaluate the effect of API on Compritol® 888 processibility.

3 - RESULTS & DISCUSSION

The first part of the investigation evaluated the process boundary of Compritol® 888 alone. The main criteria was the extrudate quality/aspect : solid & smooth strands are presented by green cross and lack of strands by red cross in Figure 1. These parameters are adapted for the extrusion of Compritol® 888 alone using a Pharma 11 extruder and the temperature profile set to 60-63-66-69-66-65-69-68°C towards a 1 mm die.

When studying feed rate and screw speed, optimal extrudates were obtained with screw speed below 200 rpm and feed rates below 0.4 kg/h. When feed rates are higher than 0.4 kg/h, die pressure and torque increased. Figure 2 shows that screw speed can reduce this effect but it is limited to 250 rpm.

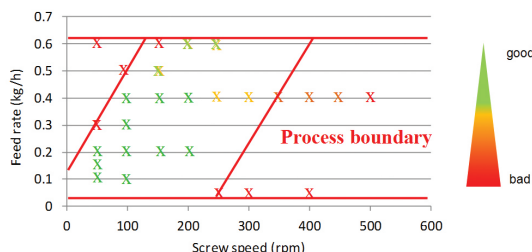


Figure 1: Potential operating window for Compritol® 888 extrusion at a temperature profile of 60-63-66-69-66-65-69-68 towards the die (1mm).

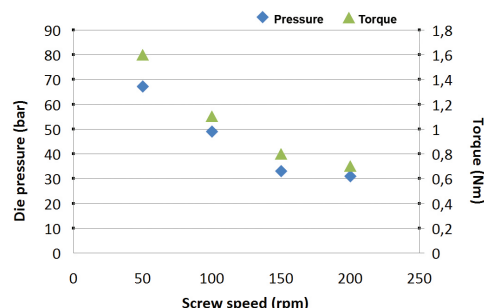


Figure 2: Impact of screw speed on die pressure and torque during Compritol® 888 extrusion at a constant feed rate of 0.2kg/h.

The significance of these two parameters was plotted in the Pareto chart shown in Figure 3a and 3b. For both graphs, it is seen that changing feed rate (F) and screw speed (S) significantly impact die pressure (3a) and torque (3b). The significance is indicated by values being above the orange limit (1.37bars for die pressure and 0.04Nm for torque). When studying barrel and die temperatures, the most critical parameter was the die temperature due to the narrow melting range temperature of Compritol® 888 (68.5 - 71.5°C).

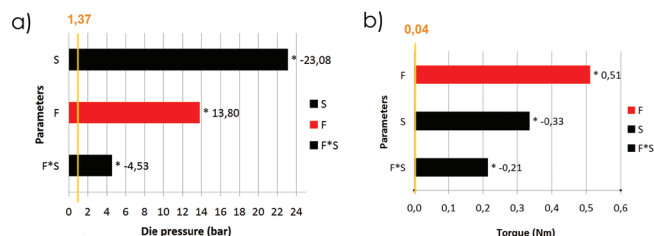


Figure 3: Significance of feed rate F and screw speed S on: a) die pressure and b) torque.

Figure 4 illustrates the temperature limit of the die as a function of the temperature of the most important heating zones (Z8, 7 and 5). The results indicate that die temperature was most critical: 68°C was the lower temperature limit and 73°C was the upper limit. These parameters are adapted for the extrusion of Compritol® 888 alone using a Pharma 11 extruder and at a feed rate of 0.2kg/h and a screw speed of 100rpm towards a 1 mm die.

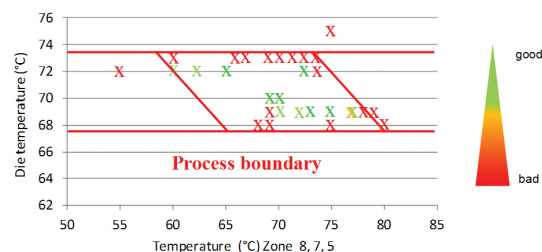


Figure 4: Potential temperature boundary for Compritol® 888 extrusion at a feed rate of 0.2 kg/h, screw speed of 100 rpm, 1 mm die.

The second part of the investigation was performed with Compritol® 888 and 10% of niacin. This shows how the addition of a water-soluble drug may influence the process boundaries. Figure 5 shows the impact of niacin on the temperature boundary. The presence of 10% of niacin provided a larger operating window. Lower temperatures were applicable for the barrels and for the die -this maybe due to a plasticizing effect of the drug [3]. It is also possible to work with higher processing temperatures which could be due to the higher viscosity of the drug/lipid melt compared to the molten lipid alone.

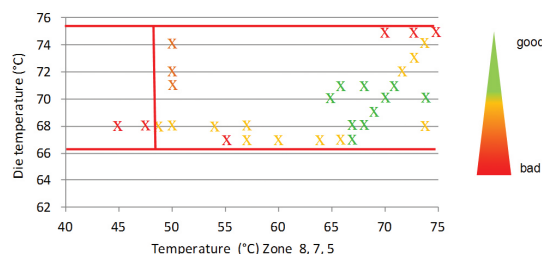


Figure 5: Effect of niacin addition (10% w/w) on extrusion temperature boundary at a feed rate of 0.2 kg/h, screw speed of 100 rpm.

The final part of this study shows the ability of Compritol® 888 to disperse hydrophilic and lipophilic drugs when processed in HME, illustrating the homogeneity of niacin particles in the Compritol® matrix. To simulate this, a hydrophilic dye was added to Compritol® 888. It is important to check that the addition of a hydrophilic water soluble drug such as niacin to a lipophilic water insoluble lipid matrix does not lead to drug rejection or uneven dispersion.

Figure 6a illustrates that the hydrophilic compound was not rejected from the lipophilic matrix and distribution was homogenous. The same experiment was done with a hydrophobic dye (Figure 6b). This demonstrates the good repartition of both hydrophilic and hydrophobic drugs in the Compritol® 888 matrix.

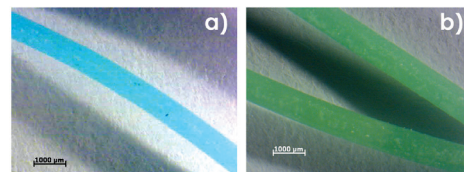


Figure 6: Simulation of drug distribution in the Compritol® 888 matrix using: a) a hydrophilic and b) a hydrophobic dye.

4 - CONCLUSION

Since HME is gaining interest as a manufacturing method for oral solid dosage forms, it is important to evaluate the process boundaries when using a lipid excipient as matrix former. This work presents a first attempt to better understand the importance of process parameters and drug incorporation on lipid extrusion. The die temperature was found to be the most crucial parameter for Compritol® 888 ATO extrusion and optimally should be within the excipient melting range. If die temperatures are too low then lipid solidification and die plugging occurs, whereas if temperatures are too high solid extrudates cannot be obtained. In this study, the addition of niacin allowed to work in a larger operating window. Hydrophilic as well as hydrophobic compounds were proven to be evenly distributed in the Compritol® 888 matrix.

5 - REFERENCES

- [1] Repka, M.A. Pharmaceutical Applications of Hot-Melt Extrusion: Part II. Drug Dev. Ind. Pharm. 33, 1043-1057 (2007).
- [2] Kleinebudde, P. Solid lipid extrusion. AAPS Advances in the Pharmaceutical Sciences 9, 299-328 (2013).
- [3] Siepmann, F. Drugs acting as plasticizers in polymeric systems: a quantitative treatment. J Control Release 115, 298-306 (2006).